



Diversifying equity with cryptocurrencies during COVID-19

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ARTICLE INFO

JEL classification:

C58

F37

G14

G15

Q31

Keywords:

Co-movement

COVID-19

Bitcoin

Wavelet

Safe haven

ABSTRACT

Literature suggests assets become more correlated during economic downturns. The COVID-19 crisis provides an unprecedented opportunity to investigate this considerably further. Further, whether cryptocurrencies provide a diversification for equities is still an unsettled issue. We employ several econometric procedures, including wavelet coherence, and neural network analyses to rigorously examine the role of COVID-19 on the paired co-movements of four cryptocurrencies, with seven equity indices (matching countries particularly impacted by COVID-19). Our period of study includes one year prior to the onset of COVID-19, and one year during the pandemic, extending deeper into the pandemic period (February 2021) than most previous studies. We find co-movements between cryptocurrencies and equity indices gradually increased as COVID-19 progressed. However, most of these co-movements are either modestly positively correlated, or minimal, suggesting cryptocurrencies in general do not provide a diversification benefit during either normal times or downturns. An exception, however, is the co-movement of tether. Tether co-moves negatively with equities to an economically significant degree, both pre COVID-19, and considerably more during COVID-19. Co-movements between tether and equity indices spiked sharply during identified waves of the pandemic. Tether appears to be an important safe haven during times of market turmoil, consistent with investors seeking USD liquidity during periods of volatility.

1. Introduction and motivations

As noted by Goodell (2020), the COVID-19 pandemic will likely impact financial systems, institutions, and markets to an unprecedented extent into the future. There are ample reasons to suspect COVID-19 will change the dynamics of financial markets, both during the pandemic, and into the future as a result of changes in attitudes and behaviors. Some even suggest that COVID-19 will alter entire research streams. For instance, Yarovaya, Brzezczynski, Goodell, Lucey, and Lau (2020a) outline how COVID-19 will necessitate a large-scale reexamination of research into financial market contagion. It is likely that even well-researched topics will warrant updating in light of COVID-19. Such research interest will naturally include re-examining how asset classes co-move, both with both COVID-19 and each other.

Indeed, a reexamination of financial market dynamics in the context of during COVID-19, and post COVID-19, is already underway. Recent papers examine the impact of COVID-19 across industries (Goodell & Huynh, 2020). Others examine impact of COVID-19 on herding behavior (Yarovaya, Matkovskyy, & Jalan, 2020b) and co-movements across commodities, equities, and economic policy uncertainty (Aloui, Goutte, Guesmi, & Hchaichi, 2020; Sharif, Aloui, & Yarovaya, 2020).

Of particular relevance to this paper, researchers are examining the correlations, hedging, diversifying and safe haven properties of cryptocurrencies with respect to how previously identified relations have changed during the COVID-19 crisis (Conlon, Corbet, & McGee, 2020; Conlon & McGee, 2020; Goodell & Goutte, 2021).

We employ several econometric procedures, including wavelet coherence analysis (Kang, McIver, & Hernandez, 2019; Goodell & Goutte, 2021), and neural network analysis to examine the role of COVID-19 on the paired co-movements of four cryptocurrencies, with seven equity indices. To provide clarity of comparison, we use similar combinations of cryptocurrencies and equity indices as the recent (Conlon et al., 2020), while removing some less liquid exchanges. Cryptocurrencies examined include bitcoin, ethereum, litecoin and tether. Equity indices include SWISS Index, IBEX 35, DAX, CAC 40, FTSE 100, EUROSTOXX, and S&P 500. In addition, we also include the CBOE VIX for further clarifying inferences regarding co-movements with volatility. Our indices reflect the markets of countries particularly impacted by COVID-19. Our period of study includes one year prior to the onset of COVID-19, and one year during the pandemic, extending deeper into the pandemic period (February 2021) than most previous studies.

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<https://doi.org/10.1016/j.irfa.2021.101781>

Received 20 June 2020; Received in revised form 6 April 2021; Accepted 20 April 2021

Available online 24 April 2021

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Goodell (2020) and others suggest that equity risk will be reevaluated during and post COVID-19. Further, literature already offers a number of studies evidencing the impact of COVID-19 on equity markets (Conlon et al., 2020; Conlon & McGee, 2020; Corbet, Larkin, & Lucey, 2020; Corbet, Lucey, Peat, & Vigne, 2018a; Goodell & Goutte, 2021). We consider the following hypothesis

H1: COVID-19 impacted the relationship between cryptocurrencies and equity markets

We also consider that the cryptocurrencies in our sample have fundamental differences amongst them. Tether for instance, being a stable coin, roughly based on the USD, is inherently quite different from bitcoin etc. Beneki, Koulis, Kyriazis, and Papadamou (2019) and Sifat, Mohamad, Bin, and Shariff (2019) consider potential differences between bitcoin and ethereum. Liu, Wan, Zhang, and Zhao (2019) suggest that bitcoin may differ from other cryptocurrencies that lack a futures market. Liu (2019), Goodell and Goutte (2021), and Griffin and Shams (2020) evidence differences between tether and other cryptocurrencies. Given likely heterogeneity amongst cryptocurrencies, it is intuitive to consider that COVID-19 will have heterogeneous impact on them. It is reasonable to consider that an event of the magnitude of COVID-19 will impact differing cryptocurrencies in differing ways. We hypothesize the following

H2: There is heterogeneity among cryptocurrencies in how COVID-19 impacted their respective relationship with equity

We also consider that tether particularly differs from other cryptocurrencies with respect to being a safe haven during economic downturns. We consider that tether will co-move with equity differently from other cryptocurrencies because it has particular utility as a vehicle for liquidity. Tether is particularly highly traded, largely because it is a stable currency pegged to the US dollar. In other words, a single tether is de jure redeemable for a single US dollar.¹ We therefore hypothesize that in times of economic crisis there is a need for tether. This is especially true during crises as it is often difficult for cryptocurrency exchanges to hold sovereign currencies, without the depositor possessing a special license. Tether provides traders an alternative to holding USD, allowing for moving between a somewhat equivalent form of USD and other cryptocurrencies without holding sovereign currency or undertaking special registration. Tether, being bought and sold through cryptocurrency exchanges can easily be transferred between parties without requiring banking intermediation. Therefore, tether is naturally highly liquid, and a convenient store of value during times of economic downturn, liquidity constraint, or highly volatile markets.

H3: Tether differs from other cryptocurrencies with respect to being a safe haven during severe economic downturns.

We find co-movements between cryptocurrencies and equity indices gradually increased as COVID-19 progressed. However, most of these co-movements are either modestly positively correlated, or minimal, suggesting cryptocurrencies in general do not provide a diversification benefit during either normal times or downturns. An exception, however, is the co-movement of tether being negative with equities, both pre COVID-19, and considerably more pronounced during COVID-19. We evidence that co-movements between tether and equity indices spiked sharply during identified waves of the pandemic. Tether appears to be an important safe haven during times of market turmoil, consistent with investors seeking USD liquidity during periods of volatility.

Additionally, we evidence other relationships that suggest the cryptocurrencies bitcoin, ethereum and litecoin are not safe havens or even diversifiers. In robustness testing, we find equity indices co-moved negatively with the VIX, as expected. However, co-movements of bitcoin, ethereum, and litecoin co-moved together negatively with the

VIX, while tether moved positively with the VIX. These results affirm the special value of tether as a safe haven. Overall, evidence suggests that most cryptocurrencies are not safe havens (consistent with Conlon et al., 2020). However, in contrast to the other cryptocurrencies, tether is a safe haven.

Within the context of COVID-19, this article is motivated in several ways. First, it is known, at least since Hartmann, Straetmans, and de Vries (2004), that assets and asset classes become more correlated during economic downturns (see also Bekaert, Hodrick, & Zhang, 2009; Lee, Lin, & Yang, 2011). Certainly COVID-19 presents an unprecedented opportunity to investigate this further.

Second, there is a great deal of literature interest in whether bitcoin, and cryptocurrencies in general, have gold-like safe-haven properties. Since Dyhrberg (2016a, 2016b), there has been interest in whether bitcoin might share some of the same hedging properties as gold (Baur & Lucey, 2010; Baur & McDermottDirk, 2010; Bouri, Lucey, & Roubaud, 2019; Bredin, Conlon, & Poti, 2015; Gil-Alana, Abakah, & Rojo, 2020; Goodell & Goutte, 2021; Klein, Thu, & Walther, 2018; O'Connor, Lucey, Batten, & Baur, 2015; Raza, Shah, & Shahbaz, 2018; Smales, 2019). The issue of whether cryptocurrencies are coupled with traditional asset classes is also not settled. Corbet, Meegan, Larkin, Lucey, and Yarovaya (2018b), Feng, Wang, and Zhang (2018), Mensi, Rehman, Maitra, Al-Yahyaee, and Sensoy (2020) and Shahzad, Bouri, Roubaud, and Kristoufek (2019) and many others find nuanced evidence correlation of cryptocurrencies with other assets. However, this issue begs reconsideration for the period of COVID-19.

Third, there is ongoing interest in how asset classes behave under uncertainty generally (e.g., Goodell, McGee, & McGroarty, 2020), and, more particularly, how cryptocurrencies behave under shocks to uncertainty (Kurka, 2019; Matkovskyy & Jalan, 2019; Wang, Xie, Wen, & Zhao, 2019; Wu, Tong, Yang, & Derbali, 2019). There is considerable early interest in how the relationships and characteristics of cryptocurrencies might be impacted by the COVID-19 crisis, both during the pandemic and extending into the future (Goodell & Goutte, 2021).

Fourth, we consider, following Hale, Krishnamurthy, Kudlyak, and Shultz (2018), that investors in tether are more specialized and informed traders than those investing in cryptocurrencies in general (da Gama Silva, Neto, & Klotzle, 2019; Wei, 2018). Wei (2018) finds that when bitcoin suffers a decline there is a subsequent increase in the volume of tether trading.

We find that positive co-movements between the cryptocurrencies bitcoin, ethereum, and litecoin with equity indices, while not particularly substantial, got progressively larger, as COVID-19 progressed, consistent with these cryptocurrencies correlating with other assets during downturns. Clearly, they are not safe havens for equity. On the other hand, we evidence a progressing negative co-movement during the increase of COVID-19 between equity indices and tether, consistent with tether acting as a genuine safe haven.

As discussed further in the paper, we interpret our findings of tether generally acting differently from other cryptocurrencies from several perspectives. These results could be reflective of different investor types (Grinblatt & Keloharju, 2020). Additionally, the results for tether could be reflective of investors moving from other cryptocurrencies toward the noted stability of tether. Our results should be of great interest to a broad spectrum of academics, and investors.

2. Data and statistics

Our data covers a roughly two-year period from the 28 February 2019 to 09 February 2021. Data is from Thomson Reuters Datastream. The motivation for choosing this two-year period is to merge, and so investigate the issue of diversification using cryptocurrencies for two different economic and financial situations. Namely,

- February 2019 to February 2020: a standard economic situation
- March 2020 to February 2021: the COVID 19 pandemic situation, with concomitant impact on financial markets

¹ We realize that it is contested whether tether is indeed de facto redeemable for the full value of a dollar

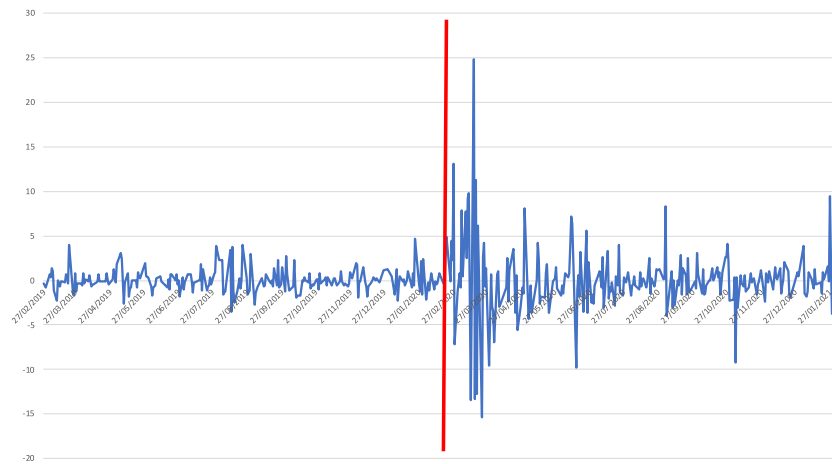


Fig. 1. VIX returns between February 2019 and February 2021.

Table 1
Descriptive statistics — Part 1.

	BTC	Litecoin	Ethereum	Tether
Missing	471	471	471	471
Mean	86.888	0.255	3.321	1.210e-5
Std. Deviation	759.166	5.326	30.594	0.002
Skewness	2.226	-0.167	0.452	0.565
Std. Error of Skewness	0.113	0.113	0.113	0.113
Kurtosis	27.056	11.258	26.641	22.650
Std. Error of Kurtosis	0.225	0.225	0.225	0.225
Shapiro-Wilk	0.674	0.844	0.629	0.721
P-value of Shapiro-Wilk	< .001	< .001	< .001	< .001
Minimum	-4664.670	-40.640	-270.170	-0.015
Maximum	6811.290	26.170	211.430	0.018

Table 2
Descriptive statistics — Part 2.

	SWISS index	IBEX 35 index	DAX index	CAC 40
Missing	471	471	471	471
Mean	2.824	-2.101	5.439	1.031
Std. Deviation	112.435	118.586	186.161	74.586
Skewness	-1.121	-0.560	-0.458	-0.734
Std. Error of Skewness	0.113	0.113	0.113	0.113
Kurtosis	10.580	4.904	4.391	5.190
Std. Error of Kurtosis	0.225	0.225	0.225	0.225
Shapiro-Wilk	0.896	0.931	0.932	0.921
P-value of Shapiro-Wilk	< .001	< .001	< .001	< .001
Minimum	-869.990	-648.600	-1004.480	-408.550
Maximum	584.510	552.600	745.310	314.390

This choice is influenced by examining Fig. 1, where we can see that after March 2020, the volatility of VIX returns increased dramatically.

We compare results for the full time period to the period of the COVID 19 pandemic to investigate changes and to investigate the impact of COVID-19 stress on diversification issues. To provide clarity of comparison, we employ similar combinations of cryptocurrencies and equity indices as Conlon et al. (2020). Cryptocurrencies examined include bitcoin, ethereum, litecoin and tether. Equity indices include SWISS Index, IBEX 35 Spain, DAX, CAC 40 French, FTSE 100 UK, EUROSTOXX and S&P 500. In addition, we also include the CBOE VIX for comparative insights. Descriptive statistics for the two sample periods are shown in the Tables 1 to 3.

2.1. Correlation analysis

Figs. 2 and 3 show the correlation matrix between each variables using heat-map representations for the full sample period and the

Table 3
Descriptive statistics — Part 3.

	FTSE 100 index	EURO STOXX 50	S&P 500	CBOE VIX
Missing	471	471	471	471
Mean	-1.332	0.819	2.384	0.013
Std. Deviation	89.779	43.621	37.757	2.850
Skewness	-1.231	-0.773	-0.728	1.014
Std. Error of Skewness	0.113	0.113	0.113	0.113
Kurtosis	9.146	9.289	3.751	18.578
Std. Error of Kurtosis	0.225	0.225	0.225	0.225
Shapiro-Wilk	0.891	0.889	0.927	0.758
P-value of Shapiro-Wilk	< .001	< .001	< .001	< .001
Minimum	-639.040	-298.890	-194.740	-15.320
Maximum	452.120	249.380	160.370	24.860

COVID-19 period respectively. A dark red color indicates that the respective two variables are extremely negatively correlated, while dark blue indicates an extremely positive correlation. For instance, we can see in both Figs. 2 and 3 that the VIX is generally negatively correlated with the cryptocurrencies and equity indices for both periods of investigation. This, of course, is expected as the VIX is a volatility market index. A result of greater interest is that tether has a negative correlation with both equities and other cryptocurrencies. This argues in favor of tether being a safe haven asset.

Moreover, we can see that this negative correlation increases for the COVID-19 period. The numbers in the respective row of Fig. 3 are more negative than the row for tether in Fig. 2. This suggests that during crises, and times of volatility and stress in financial markets, the ability of tether to act as a safe haven asset is enhanced. Given Bekaert et al. (2009) and others findings that, generally, diversified asset classes become more correlated during economic downturns, this negative co-movement property of tether is very relevant for portfolio planning.

We see too that during the pandemic period the correlation between the other three cryptocurrencies in our sample, namely bitcoin, ethereum and litecoin (i.e., those cryptocurrencies other than tether), increases. This evidence, together with positive correlation of these three cryptocurrencies, suggests that during normal times, and especially in periods of high market stress, these three cryptocurrencies have little utility as diversifiers, let alone safe havens. These results are consistent with Conlon et al. (2020), Conlon and McGee (2020) and Goodell and Goutte (2021).

3. Neural network analysis

We further investigate the co-movements of cryptocurrencies, and equities with neural network analysis. This methodology allows us to

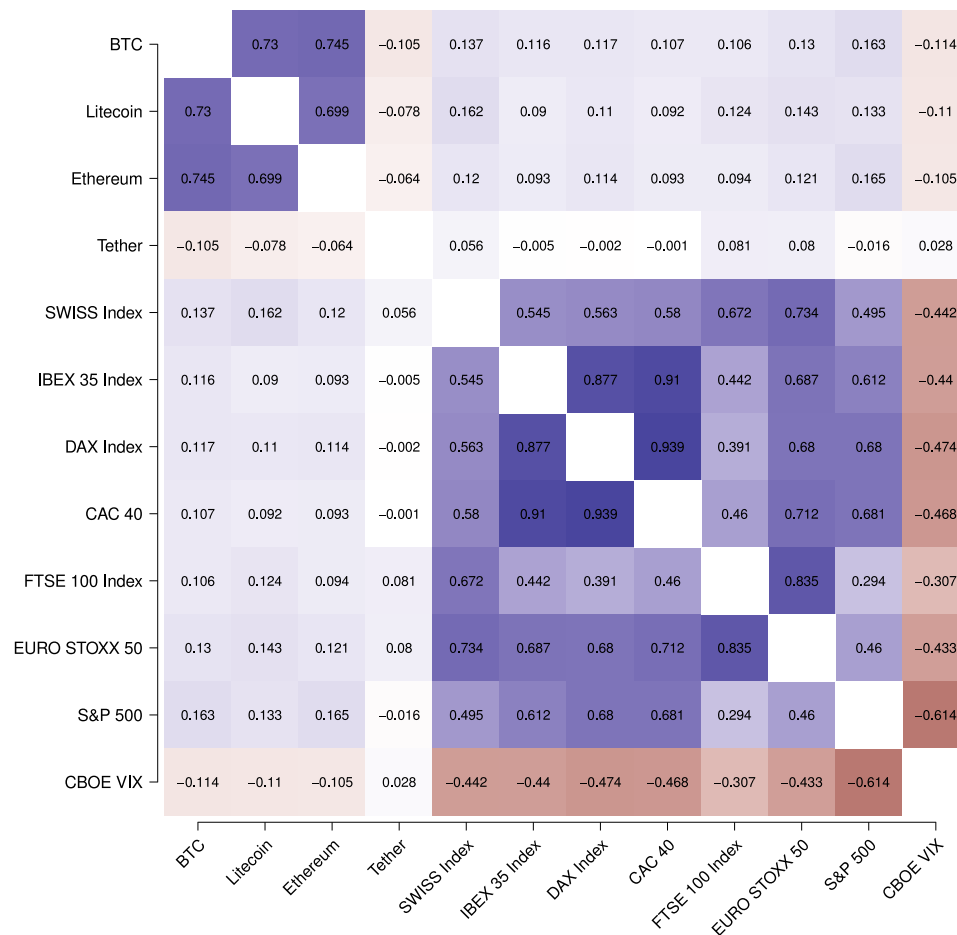


Fig. 2. Heatmap of the return series for the full period.

identify the connections and nodes between our variables. Our neural network analysis is shown in Figs. 4 and 5 for the both time periods of investigation. In these figures, a wider or thicker line between two variables indicates greater connection between them. This suggests higher dependency between these variables; and so greater potential impact and co-movements between them. Especially valuable, neural network analysis also captures the sign of the relationship, and so the nature of the correlation. In the figures, a blue connecting line denotes a positive relationship (co-moving in the same direction; positively correlated). On the other hand, a red connecting line denotes a negative correlation. And so, in the context of our analysis, a red connecting line between a cryptocurrency and an equity index indicates that this respective cryptocurrency is a diversifier, or even a safe haven, with respect to this respective equity market.

Examining Figs. 4 and 5, it is clear that all cryptocurrencies, with the exception of tether, are bunched together on the right-hand side of the network with positive (blue-colored) relationships. These results suggest that, with the exception of tether, cryptocurrencies are not diversifiers, much less safe havens, during both normal times and during extreme downturns.

Further, neural network analysis confirms our correlation analysis with regard to tether. Examining the figures for our neural network analysis, tether has no direct connection nodes with the other cryptocurrencies. This indicates considerable differences in the specificities and characteristics of tether compared to other cryptocurrencies. Tether has a behavior, relationship, and impact on the other assets (in this case equities) that is different from other cryptocurrencies. And looking Tables 4 and 5, we can see that this result is stable for both the full period and the pandemic situation, with the special behavior of tether more pronounced during the pandemic period.

Additional examination of the color of the neural network links for the pandemic period (Fig. 5) allows other inferences. For instance, we see that tether has a red connection with the S&P 500. It also has a positive relationship with the VIX. Together these results suggest tether has the role of a safe-haven asset with respect to US equities during the crisis period.

Interestingly, neural network analysis also shows a negative or null relationship between tether and other cryptocurrencies for both periods of investigation (pre-crisis and during the crisis). This suggests that tether plays a different role in portfolio diversification than bitcoin, litecoin and ethereum. We evidence that tether is not only a diversifier for equity but also for bitcoin, litecoin, and ethereum investors. In contrast, bitcoin, ethereum and litecoin positively link with each other.

Further, of tangential interest to our main focus, amongst bitcoin, litecoin, and ethereum, we see that bitcoin plays a major role over the full period, leading the movements of litecoin and ethereum. Indeed, we can see that the main important connection in the non-tether cryptocurrency group comes from bitcoin impacting ethereum (0.471) and then further dispatching to litecoin and ethereum. For the pandemic period, this pattern of bitcoin leading litecoin and ethereum stays true, but with litecoin increasing its connectivity to both ethereum and bitcoin. These various results confirm that in periods of high market volatility the connectivity of the cryptocurrencies other than tether (bitcoin, ethereum, and litecoin) becomes even closer than it is during normal market circumstances. These results, along with the separation of bitcoin, ethereum, and litecoin from the equity indices (as clearly evident in our figures) suggests that these three cryptocurrencies are clearly not safe havens either during normal times or particularly during times of market turmoil.

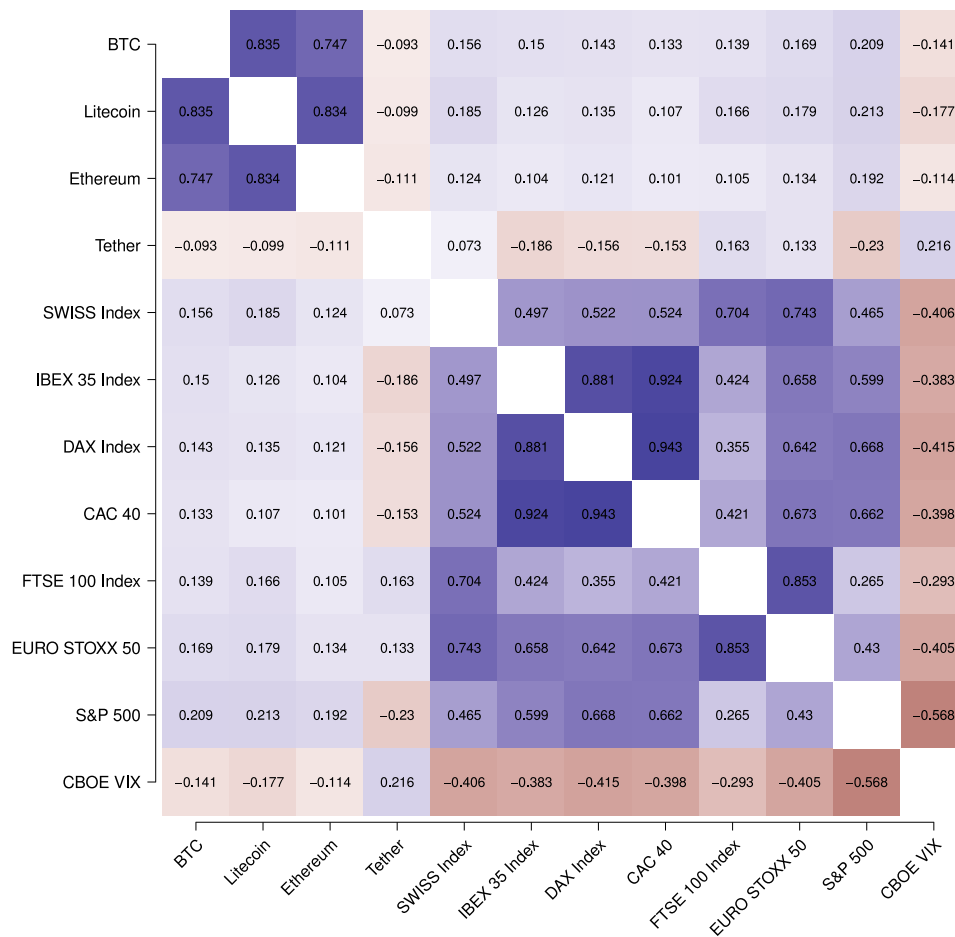


Fig. 3. Heatmap of the return series for pandemic period.

Table 4

Weights matrix of our Neural network for the full period.

Variable	Network											
	bitcoin	litecoin	ethereum	tether	SWISS	IBEX 35	DAX	CAC 40	FTSE 100	EUROSTOXX	S&P 500	VIX
bitcoin	0.000	0.430	0.471	-0.058	0.000	0.002	0.000	0.000	0.000	0.003	0.024	0.000
litecoin	0.430	0.000	0.338	-0.006	0.042	-3.040e-5	0.000	-0.012	0.000	0.017	0.000	0.000
ethereum	0.471	0.338	0.000	0.000	0.000	0.000	0.000	-0.005	0.000	0.000	0.042	0.000
Tether	-0.058	-0.006	0.000	0.000	0.013	-0.019	0.000	-0.006	0.006	0.050	0.000	0.032
SWISS	0.000	0.042	0.000	0.013	0.000	0.000	0.050	0.000	0.228	0.203	0.130	-0.093
IBEX 35	0.002	-3.040e-5	0.000	-0.019	0.000	0.000	0.178	0.468	0.000	0.078	0.000	0.000
DAX	0.000	0.000	0.000	0.000	0.050	0.178	0.000	0.608	-0.200	0.165	0.141	0.000
CAC 40	0.000	-0.012	-0.005	-0.006	0.000	0.468	0.608	0.000	0.000	0.112	0.139	0.000
FTSE 100	0.000	0.000	0.000	0.006	0.228	0.000	-0.200	0.000	0.000	0.713	-0.010	0.000
EUROSTOXX	0.003	0.017	0.000	0.050	0.203	0.078	0.165	0.112	0.713	0.000	-0.093	-0.058
S&P 500	0.024	0.000	0.042	0.000	0.130	0.000	0.141	0.139	-0.010	-0.093	0.000	-0.415
VIX	0.000	0.000	0.000	0.032	-0.093	0.000	0.000	0.000	0.000	-0.058	-0.415	0.000

4. Wavelet coherence analysis

In this section, we undertake wavelet coherence analysis. We apply the wavelet methodology of [Grinsted, Moore, and Jevrejeva \(2004\)](#), recently used by [Goodell and Goutte \(2021\)](#) and others, to further study the relationship between cryptocurrencies and equity during COVID-19.

[Figs. 6 and 7](#) illustrate this analysis. In these illustrations, coherency ranges from yellow (high coherency) to blue (low coherency) to designate the amount of co-movement. A yellow color represents strong co-movement(s), while a blue color indicates weak co-movement(s). Additionally, we discern causality and phase differences. Arrows indicate the phase differences between the two assets. For example, $\rightarrow$ and $\leftarrow$ indicate that the assets are in phase and out of phase respectively.

Being in phase (out of phase) indicates a positive (negative) correlation between respective variables. Moreover, $\nearrow$ and $\searrow$ indicate that the first asset is leading the second one, while $\nwarrow$ and $\swarrow$ indicate the first asset values are lagging those of the second one. We initially focus on the co-movement of tether, as we have identified with heatmap and neural network analysis that the co-movement of tether with equity, in contrast to bitcoin, ethereum, and litecoin, is negative, consistent with being a safe haven.

Examining [Fig. 6](#), we clearly see large swathes of yellow-colored areas of high coherency. Further, the arrows are predominantly left-pointing, consistent with being out of phase or indicating negative co-movement. This behavior is consistent across the co-movements of tether with each of the equity indices.

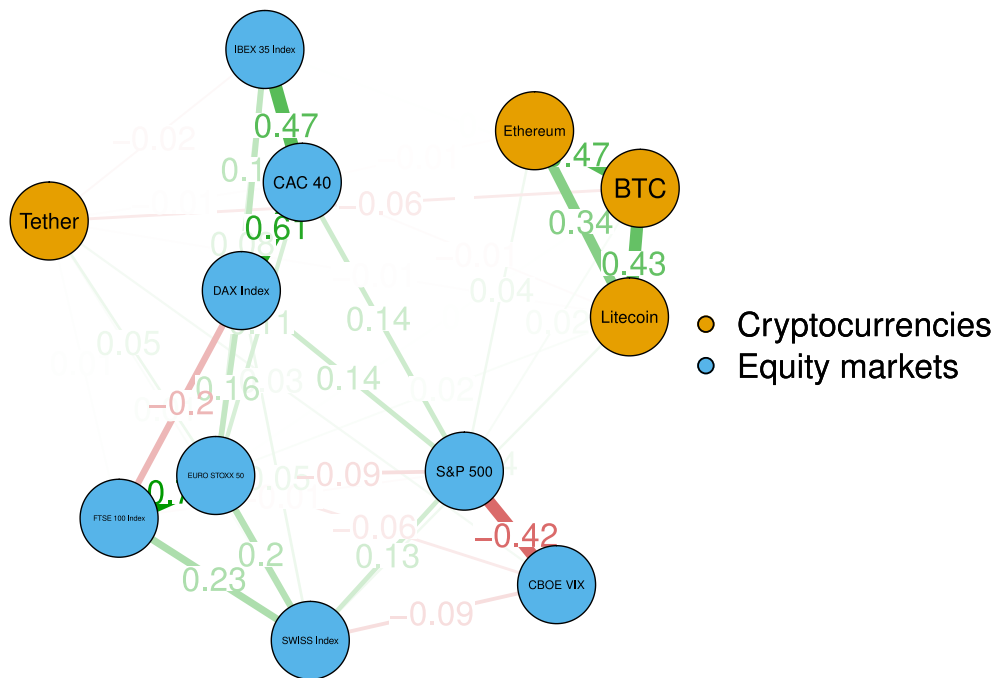


Fig. 4. Neural network map for the full period.

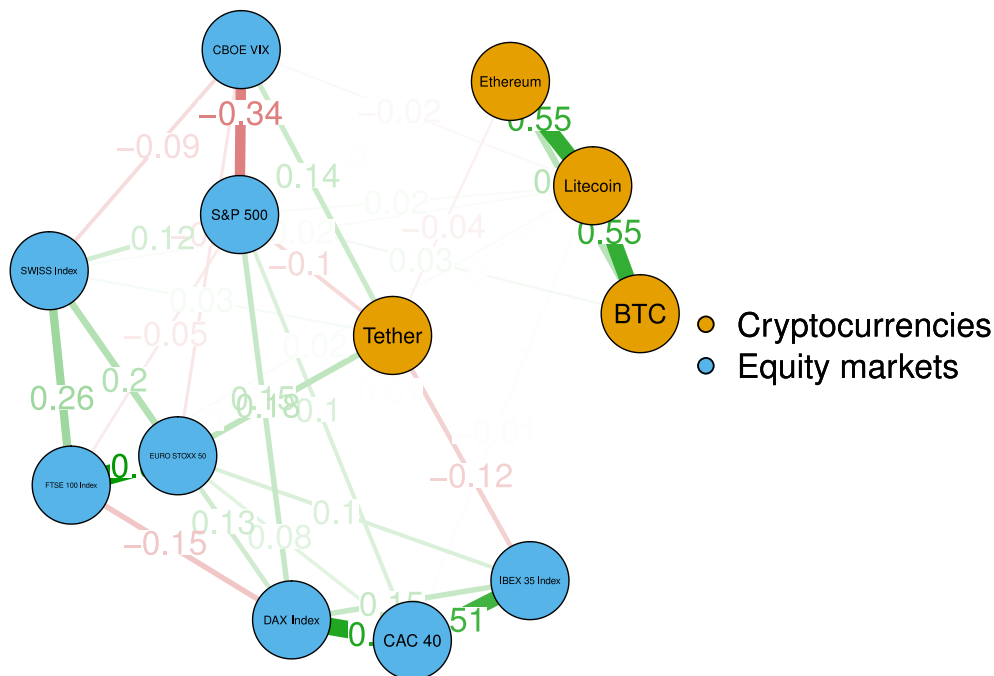


Fig. 5. Neural network map for the pandemic period.

Following on, Fig. 7, illustrates the co-movements of the four cryptocurrencies with the S&P500. We select the S&P500 as representative for brevity. The co-movements of the cryptocurrencies with the other equity indices in our sample are qualitatively the same. Examining Fig. 7, we clearly see that there are co-movements of bitcoin, ethereum, and litecoin with S&P500, as show by the significant yellowed areas. Further, the arrows on the diagrams for bitcoin, ethereum, and litecoin, in contrast to tether, generally point to the right. This indicates positive, in-phase co-movement. Therefore, Fig. 7 evidences that tether is a diversifier for the S&P 500, while bitcoin, ethereum, and litecoin are not.

4.1. Significant increasing of contagion during waves of COVID-19

Fig. 8 shows a plot of the dynamic contagion parameter between tether and the S&P 500. Greater height on this graph indicates a higher level of contagion between tether and the S&P 500. Clearly visible is that contagion spikes for periods corresponding to waves of COVID-19 (listed below).

- February to April 2020
- July to August 2020
- November to December 2020
- February 2021

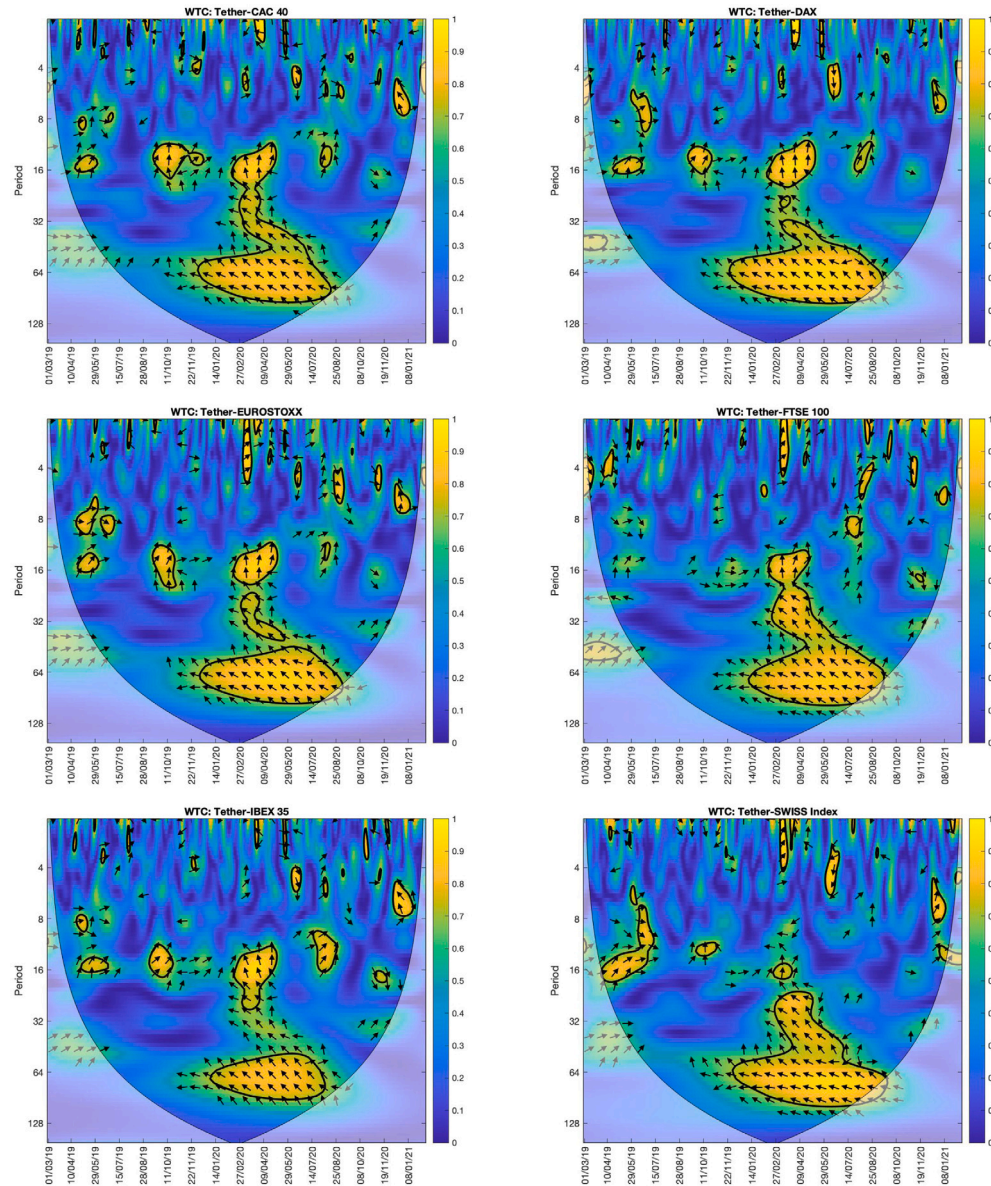


Fig. 6. Wavelet coherence (WTC) between tether and equities. The level of correlation is indicated by the color on the right side of the charts; the hotter the color, indicated by yellow versus blue, the higher the absolute correlation value with respect to $R^2(u, s)$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Weights matrix of our Neural network for the pandemic period.

Variable	c											
BTC	0.000	0.551	0.190	0.000	0.000	0.000	0.000	0.000	0.000	0.005	0.033	0.000
Litecoin	0.551	0.000	0.552	-0.003	0.017	0.000	0.000	-0.009	0.015	0.005	0.020	-0.018
Ethereum	0.190	0.552	0.000	-0.039	0.000	0.000	0.000	0.000	0.000	0.000	0.005	0.000
Tether	0.000	-0.003	-0.039	0.000	0.027	-0.124	0.000	0.000	0.004	0.176	-0.101	0.142
SWISS Index	0.000	0.017	0.000	0.027	0.000	0.000	0.042	0.000	0.261	0.204	0.122	-0.094
IBEX 35	0.000	0.000	0.000	-0.124	0.000	0.000	0.153	0.515	0.000	0.100	0.000	0.000
DAX	0.000	0.000	0.000	0.000	0.042	0.153	0.000	0.601	-0.154	0.128	0.152	0.000
CAC 40	0.000	-0.009	0.000	0.000	0.000	0.515	0.601	0.000	0.000	0.082	0.099	0.000
FTSE 100	0.000	0.015	0.000	0.004	0.261	0.000	-0.154	0.000	0.000	0.688	-0.055	0.000
EUROSTOXX	0.005	0.005	0.000	0.176	0.204	0.100	0.128	0.082	0.688	0.000	0.000	-0.065
S&P 500	0.033	0.020	0.005	-0.101	0.122	0.000	0.152	0.099	-0.055	0.000	0.000	-0.342
VIX	0.000	-0.018	0.000	0.142	-0.094	0.000	0.000	0.000	0.000	-0.065	-0.342	0.000

The largest spike occurs particularly for the first sub-period. However the spikes are pronounced for all waves. This figure, highlighting

strikingly strong interplay between equity and tether, suggests the our results evidencing negative co-movement between tether and equity,

are not just statistically significant, but also economically significant. This affirms that tether particularly acts as a diversifier and safe haven during periods of market turmoil.

4.2. Robustness testing: Cryptocurrencies and equity against VIX

As a robustness test, we compare the co-movement of the cryptocurrencies and equity indices in our sample with movements in the VIX. This helps to better understand our findings regarding the co-movements of cryptocurrencies with equities during COVID-19. As the VIX is generally considered an index of market volatility, reflecting market stress and uncertainty, we expect that equities will negatively co-move with the VIX (see the heatmaps on Figs. 2 and 3). Further, from a different perspective, examining our results alternatively with regard to VIX, rather than levels of COVID-19, allows for inferring that it is indeed market uncertainty that is driving the co-movements we observe.

Examining Fig. 9, which plots co-movements of each cryptocurrency with VIX, we see that the start of the pandemic is associated with an increasing and significant relationship between cryptocurrencies and VIX. In all cases, the co-movements appear only after the start of the pandemic. Further, with the exception of tether, all co-movements of cryptocurrencies with VIX are out of phase (left arrow), suggesting negative correlation. These results argue against the efficacy of bitcoin, ethereum, or litecoin as diversifiers during economic downturns, and affirm the exceptional nature of tether. These results also affirm that our results regarding co-movements during COVID-19 are due to COVID-19's role in establishing market uncertainty.

The upper left diagram in Fig. 9 illustrates specifically the co-movement of tether with VIX during COVID-19. We see that during the COVID-19 pandemic these co-movements are significant, with a lag of one week to two months, and always positively correlated (right arrows). This result strongly suggests a role for tether as a diversifier or safe haven during extreme economic downturns.

The results of wavelet coherence analysis of the cryptocurrencies in our sample with the VIX; as well as similarly comparing the equity indices of our sample with the VIX are consistent with what we find for our cryptocurrencies and equity indices against COVID-19. This suggests that our results for the pandemic situation have broader implications for market turmoil in general, and are not specific only to the COVID-19 pandemic. Further, results once again strongly suggest a role for tether as diversifier and safe haven during extreme economic downturns.

5. Discussion

Regarding our three hypotheses,

- H1: that COVID-19 impacted the relationship between cryptocurrencies and equity markets;
- H2: that there is heterogeneity among cryptocurrencies in how COVID-19 impacted their respective relationship with equity; and
- H3: that tether differs from other cryptocurrencies with respect to being a safe haven during severe economic downturns

our study provides considerable evidence. First, we find that COVID-19 impacted the relationship between equity and cryptocurrencies in a rather specific way, by increasing the economic significance of negative co-movements between tether and equity. Of more modest interest, regarding bitcoin, litecoin, and ethereum, we find some changes in the interactions amongst these three cryptocurrencies as a result of COVID-19. But we identify little role of these three cryptocurrencies in providing diversification for equity both during the normal time of one year preceding COVID-19 and during COVID-19. And so, regarding our first two hypotheses, we do find that COVID-19 impacted the relationship(s) between cryptocurrencies and equity heterogeneously.

But the economically meaningful results in these regards has mostly to do with tether expanding its role as a diversifier or even a safe haven during COVID-19.

Particularly, regarding our third hypothesis we evidence with heatmap correlation, neural network, and wavelet coherence analyses that tether behaves very differently from bitcoin, ethereum, or litecoin. The latter three cryptocurrencies offer little diversification benefit for equity investors. On the other hand, tether, while displaying little connection with the other cryptocurrencies, manifests pronounced negative co-movements with equity indices, both in normal times, and especially during the COVID-19 period. Indeed we evidence spiking of contagion between tether and equity indices for every major COVID-19 wave.

We interpret our results for tether as due to this cryptocurrency having particular utility as an instrument of liquidity. Tether is particularly highly traded, largely because it is a stable currency, pegged to the US dollar. With a single tether roughly redeemable for a single US dollar, we consider that in times of economic crisis tether offers a channel for US dollar liquidity for some investors. This is especially likely as it is often difficult for cryptocurrency exchanges to hold sovereign currency without also having a special license. Tether provides traders an alternative, allowing for moving between a somewhat equivalent form of USD and other cryptocurrencies without a holding sovereign currency or securing a necessary license or registration. Tether, being bought and sold through cryptocurrency exchanges, can easily be transferred between parties without requiring a banking intermediary. Therefore, tether is naturally highly liquid and a convenient store of value during times of economic downturn, liquidity constraint, or highly volatile markets.

That tether behaves differently because of its stable properties has been previously identified, particularly by Liu (2019). Liu (2019) finds that tether is either uncorrelated or negatively correlated with other cryptocurrencies. This characteristic of tether is also confirmed by Goodell and Goutte (2021). However, the diversification benefit of tether has not been specifically examined for severe conditions such as that of COVID-19. Further, tether has not been previously highlighted for its pronounced safe haven property during crises.

6. Conclusions

As noted by Goodell (2020) and others (e.g., Yarovaia et al., 2020a), COVID-19, because of its scope and magnitude, calls for re-examination of financial market dynamics. As it is generally accepted that most assets and asset classes become more correlated during economic downturns (e.g., Bekaert et al., 2009), COVID-19 presents an unprecedented opportunity to investigate this further. Along with this, there has been ongoing, pre-COVID-19, a great deal of interest in whether cryptocurrencies have gold-like safe-haven properties. Further, there is considerable interest in how asset classes behave under uncertainty generally and, in particular, how cryptocurrencies behave under shocks to uncertainty. For all of these reasons, our investigation of the co-movements of cryptocurrencies and equity indices is of timely importance.

Further, motivation for our study also extends past these significant questions. In this paper, we are also interested in the heterogeneity of cryptocurrencies, with respect to how they co-move with equities during both normal times and during COVID-19.

There is now ample research (Goodell & Goutte, 2021; Griffin & Shams, 2020; Liu, 2019) to expect that differing cryptocurrencies will co-move with equity in differing manners. Additionally, there is the question of whether differing cryptocurrencies react, in their co-movements, in differing ways to extreme market turmoil. Research has noted that tether behaves differently from other cryptocurrencies. We examine this question. Does tether co-move with equity differently from other cryptocurrencies? Are any such identified differences magnified during times of enhanced market volatility? COVID-19 presents a valuable opportunity to investigate these questions as well. We also

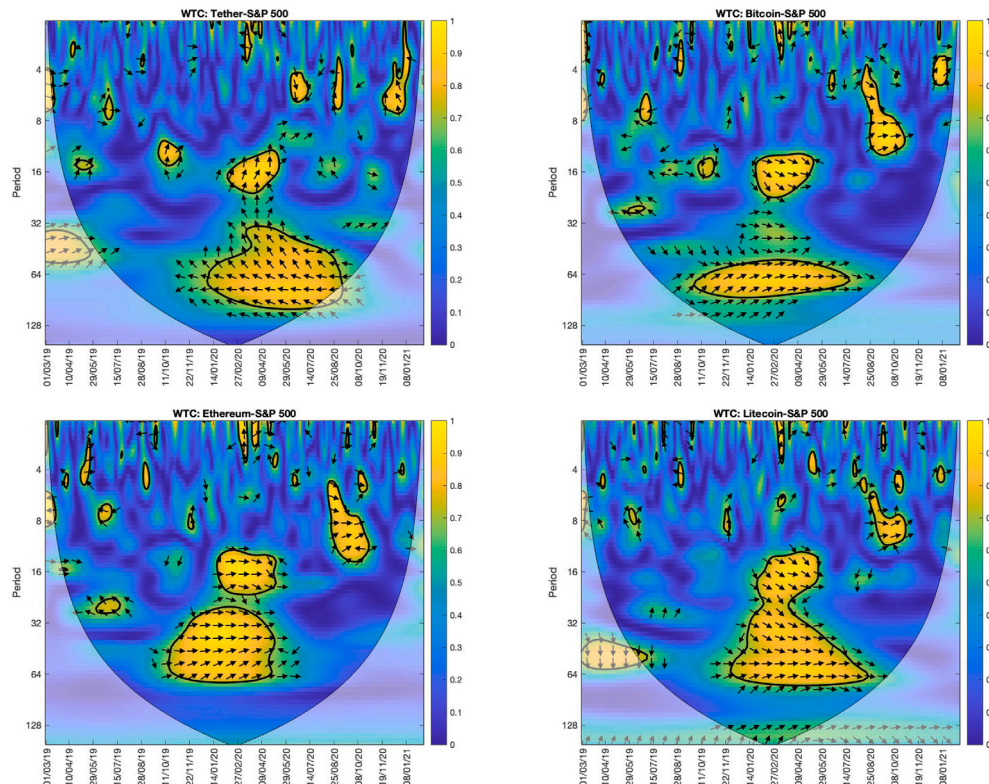


Fig. 7. Wavelet coherency (WTC) between cryptocurrencies and the S&P 500. The level of correlation is indicated by the color on the right side of the charts; the hotter the color, indicated by yellow versus blue, the higher the absolute correlation value with respect to $R^2(u, s)$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

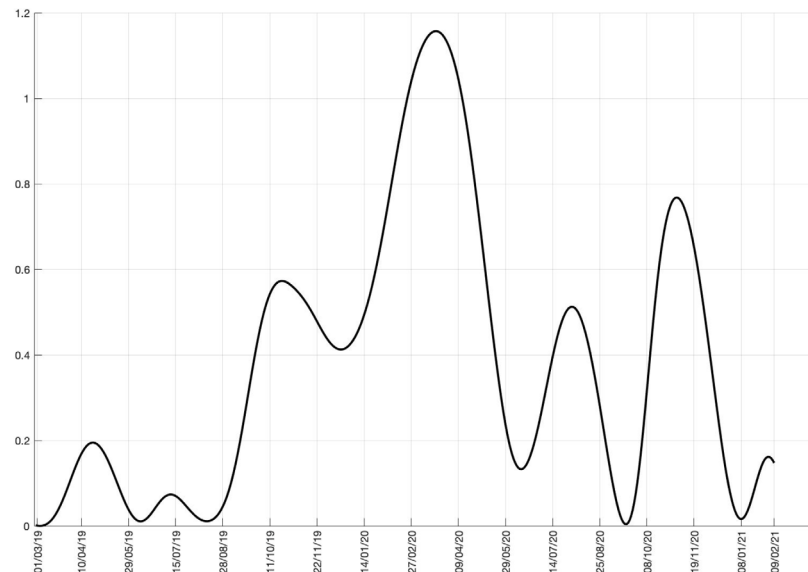


Fig. 8. Contagion parameters between tether and S&P 500.

consider that with the World on the cusp of blockchain based digital currencies replacing traditional sovereign currencies, that an investigation of the properties of tether, a cryptocurrency structured to reflect and substitute for the USD, is of especially timely importance.

We employ several econometric procedures including wavelet coherence, and neural network analyses to examine the paired co-movements of four cryptocurrencies with seven equity indices. Generally, co-movements are positively correlated, arguing against cryptocurrencies being safe havens. A notable exception, however,

is tether. The co-movements of tether are negatively correlated with equity indices, consistent with this cryptocurrency having a very different role in portfolios. Further, all co-movements between tether and equity indices got progressively larger, as COVID-19 progressed, and are economically meaningful. Indeed, we evidence that contagion between tether and equity spiked during successful waves of COVID-19 severity.

In robustness testing we also compare how the co-movements of tether with the VIX differ from the co-movements of bitcoin, ethereum,

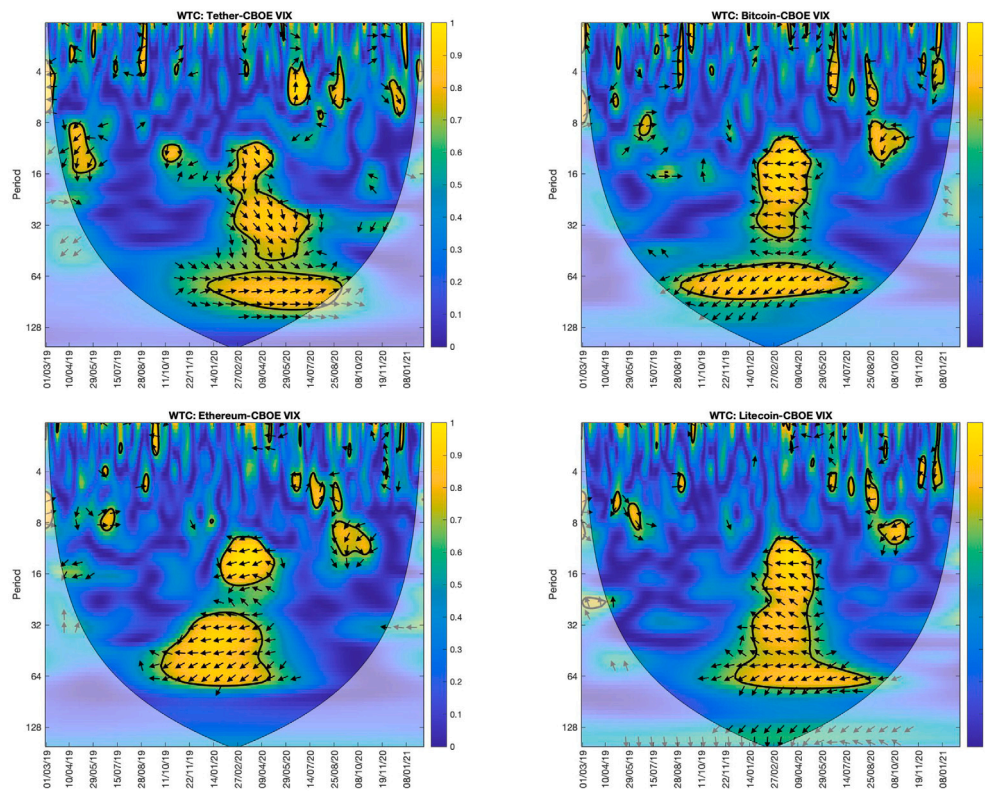


Fig. 9. Wavelet coherence (WTC) between cryptocurrencies and VIX. The level of correlation is indicated by the color on the right side of the charts; the hotter the color, indicated by yellow versus blue, the higher the absolute correlation value with respect to $R^2(u, s)$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and bitcoin. The latter three cryptocurrencies co-move slightly negatively with the VIX. However, in contrast to most cryptocurrencies, we find that tether moves positively with COVID-19 and the VIX, affirming its value as a safe haven and suggesting that the results we evidence vis-vis COVID-19 are likely due to economic uncertainty and not some other factor. In summary, while most cryptocurrencies are not safe havens, tether is. And its property as a safe haven becomes more pronounced during extreme downturns.

We interpret our findings as consistent with tether acting differently from other cryptocurrencies. This occurs in general because of differing investor types. But the primary driver of tether's quite different behavior is likely to do with its stable-coin quality of connection to the US dollar. Investors in tether are attracted toward its noted stability and structural connection to the US dollar. Our results should be of great interest to a broad spectrum of academics, and investors.

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